

$$f(x) = \sin x \ln(\sin x)$$

$$\min_{x \in (0, \frac{\pi}{2}]} f(x)$$

$$f'(x) = \cos x \ln(\sin x) + \frac{\cos x}{\sin x} \sin x$$

$$f'(x) = \cos x \ln(\sin x) + \cos x$$

$$\cos x (\ln(\sin x) + 1) = 0$$

$$\cos x = 0 \text{ or } \ln(\sin x) + 1 = 0$$

$$\cos x \neq 0 \forall x \in (0, \frac{\pi}{2}]$$

$$\sin x = \frac{1}{e}$$

$$x = \arcsin\left(\frac{1}{e}\right)$$

$$f\left(\arcsin\left(\frac{1}{e}\right)\right) = \sin\left(\arcsin\left(\frac{1}{e}\right)\right) \ln\left(\sin\left(\arcsin\left(\frac{1}{e}\right)\right)\right)$$

$$f\left(\arcsin\left(\frac{1}{e}\right)\right) = \frac{1}{e}(-) \ln(e)$$

$$f\left(\arcsin\left(\frac{1}{e}\right)\right) = -\frac{1}{e}$$